

Chemical equilibrium and routes to K

After midterm · Session 5 of 6 · 180 minutes

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The question for today

Why can composition change while the equilibrium constant stays the same?

Write your prediction before opening the lab. By the end, support an answer with one controlled comparison and an independent check.

The 180-minute class

ACTIVITY	MINUTES
Opening question and essential concepts	30
One hand calculation	20
One lab demonstration	20
Break	10
Student pair investigation	70
Discussion and exit question	30

The 70-minute investigation is student work with instructor support.

K and Q have different roles

$$\ln K = -\frac{\Delta_r G^\circ}{RT}, \quad Q = \prod_i a_i^{\nu_i}$$

K is tied to T and the fixed reaction/standard-state convention. Q describes the supplied state.

Q=K is an equilibrium condition, not an identity for every measured composition.

Several routes to standard Gibbs energy

1. Supplied $\Delta_r G^\circ$ at the evaluation T.
2. Formation properties: $\Delta_r G^\circ = \sum \nu_i \Delta_f G_i^\circ$.
3. Reaction enthalpy and entropy: $\Delta_r G^\circ = \Delta_r H^\circ - T \Delta_r S^\circ$.
4. Reference K with integrated temperature dependence.
5. Equilibrium activities, only when equilibrium is supported.

Independent inputs can disagree; the lab preserves the disagreement.

A reference-state check

Lab 12 synthetic $A_2 \rightleftharpoons 2A$ at 350 K uses $\Delta_r G^\circ = -4.000$ kJ/mol.

$$\ln K = \frac{4000}{R(350)} = 1.37454, \quad K \approx 3.95326$$

With $\Delta_r H^\circ = 50.000$ kJ/mol and $\Delta_r S^\circ = 54000/350$ J mol⁻¹ K⁻¹, the H-TS route gives the same value.

An independent check

Keep the reaction basis and standard states fixed. Convert kJ to J before calculating $\ln K$, then check $\ln Q - \ln K$ for the calculated equilibrium state.

One demonstration • 20 minutes

Open the core lab

Use the synthetic Lab 12 case at its reference temperature. Compare the supplied Gibbs-energy and H-TS routes, then change only pressure and inspect K and extent.

Keep one saved baseline for the pair investigation.

Pair investigation • 70 minutes

- 10 min: predict and record the baseline.
- 25 min: change one input and calculate.
- 20 min: exchange roles and check the result by hand.
- 15 min: prepare one plot or table and a short explanation.

In Lab 12, compare the supplied Gibbs-energy and H–TS routes at the reference temperature. Change pressure at fixed T and compare K with equilibrium extent. Check one balance.

Discussion and exit question • 30 minutes

Use 20 minutes to compare results and discuss unexpected behavior.

Why can composition change while the equilibrium constant stays the same?

Use the final 10 minutes to explain your answer individually and name one assumption that limits it.

Required practice • 45–60 minutes

In Lab 12, compare the supplied Gibbs-energy and H–TS routes at the reference temperature. Change pressure at fixed T and compare K with equilibrium extent. Check one balance.

Assessment 5 and independent checkpoint

Retain one study JSON, one comparison and a short explanation. Continue from your classroom case. The hand checkpoint is part of this time budget.

Optional study

Choose one extension: vary temperature with and without a heat-capacity correction, reverse the reaction, or scale its coefficients consistently.

[Session appendix](#) · [Module reference](#)

[Learning path](#) · [Offline package](#)

Optional work is additional. Complete the required investigation first.